

§ 2.1

1. 求下列函数的 Laplace 变换, 并给出其收敛域

1) $f(t) = \sin \frac{t}{2}$

$$F(s) = \int_0^{+\infty} \sin \frac{t}{2} e^{-st} dt$$

$$= \frac{2}{s^2 + 1}$$

$\text{Re}(s) > 0$

2) $f(t) = t^2$

$$\int_0^{+\infty} t^2 e^{-st} dt$$

$$= \frac{2}{s^3} \quad \text{Re}(s) > 0$$

2. 求下列函数的 Laplace 变换

$$f(t) = \begin{cases} 3 & 0 \leq t < 2 \\ -1 & 2 \leq t < 4 \\ 0 & 4 \leq t \end{cases}$$

$$\int_0^{+\infty} f(t) e^{-st} dt = \int_0^2 3e^{-st} dt + \int_2^4 -e^{-st} dt + \int_4^{+\infty} 0 \cdot e^{-st} dt$$

$$= \frac{1}{s} [3e^{-st}]_0^2 - \frac{1}{s} [-e^{-st}]_2^4 + 0$$

$$= -\frac{1}{s} [3e^{-2s} - 3 + -e^{-4s} + e^{-2s}]$$

$$= -\frac{1}{s} [-e^{-4s} + 4e^{-2s} - 3]$$

$$f(t) = \cos t \cdot \delta(t) - \sin t \cdot u(t)$$

$$\int_0^{+\infty} \cos t \cdot \delta(t) e^{-st} dt - \int_0^{+\infty} \sin t \cdot u(t) e^{-st} dt$$

$$= \cos t e^{-st} \Big|_{t=0} - \int_0^{+\infty} \sin t e^{-st} dt$$

$$= 1 - \frac{1}{s^2 + 1} = \frac{s^2}{s^2 + 1}$$

1) $f(t) = e^{2t}$

$$F(s) = \int_0^{+\infty} e^{2t} e^{-st} dt$$

$$= \int_0^{+\infty} e^{-(s-2)t} dt \quad \text{Re}(s) > 2$$

$$= \frac{1}{-(s-2)} [e^{-(s-2)t}]_0^{+\infty} = \frac{1}{s-2}$$

1) $f(t) = \sin t \cos t$

$$\int_0^{+\infty} \sin t \cos t e^{-st} dt$$

$$= \frac{1}{2} \int_0^{+\infty} \sin 2t e^{-st} dt$$

$$= \frac{1}{2} \cdot \frac{2}{s^2 + 4}$$

$$= \frac{1}{s^2 + 4} \quad \text{Re}(s) > 0$$

$f(t) = e^{2t} + 5 \delta(t)$

$$\int_0^{+\infty} [e^{2t} + 5 \delta(t)] e^{-st} dt$$

$$= \frac{1}{s-2} + 5(e^{-st}) \Big|_{t=0}$$

$$= \frac{1}{s-2} + 5$$

$$= \frac{5s-9}{s-2}$$

§2.2

求下列函数的 Laplace 变换

$$f(t) = t^2 + 3t + 2$$

$$F(s) = \frac{2!}{s^3} + 3 \cdot \frac{1!}{s^2} + 2 \cdot \frac{1}{s} \\ = \frac{2}{s^3} + \frac{3}{s^2} + \frac{2}{s}$$

$$f(t) = 1 - te^t \\ F(s) = \frac{1}{s} - \frac{1}{(s-1)^2}$$

$$f(t) = (t-1)^2 e^t$$

$$= t^2 e^t - 2te^t + e^t$$

$$F(s) = \frac{2!}{(s-1)^3} - 2 \cdot \frac{1!}{(s-1)^2} + \frac{1}{s-1}$$

$$f(t) = \frac{t}{2a} \sin at \\ = \frac{t}{2a} \cdot \frac{e^{jat} - e^{-jat}}{2j}$$

$$F(s) = \frac{1}{4aj} \left(\frac{1}{(s-ja)^2} - \frac{1}{(s+ja)^2} \right) \\ = \frac{1}{4aj} \cdot \frac{4ajs}{(s^2+a^2)^2} = \frac{s}{(s^2+a^2)^2}$$

若 $L[f(t)] = F(s)$, a 为实常数, 证明 (象函数的微分性质)

$$F^{(n)}(s) = (-1)^n L[t^n f(t)] \quad \operatorname{Re}(s) > c$$

特别 $L[t f(t)] = -F'(s)$, 或 $f(t) = -\frac{1}{t} L^{-1}[F'(s)]$, 并利用此结论, 计算下列各式

1) $f(t) = t e^{-st} \sin 2t$

$$L[e^{-st} \sin 2t] = \frac{2}{(s+3)^2 + 4}$$

$$(-1)^n L[t^n f(t)] = L[t^n f(t)]$$

$$= \int_0^{+\infty} (t)^n f(t) e^{-st} dt$$

$$= \int_0^{+\infty} \frac{d^n}{ds^n} [f(t) e^{-st}] dt$$

$$= \frac{d^n}{ds^n} \int_0^{+\infty} f(t) e^{-st} dt$$

$$= F^{(n)}(s)$$

$$F'(s) = -L[t f(t)] \Rightarrow L[t f(t)] = -F'(s)$$

$$f(t) = -\frac{1}{t} L^{-1}[F'(s)]$$

$$L[t e^{-st} \sin 2t] = -\frac{d}{ds} \left[\frac{2}{(s+3)^2 + 4} \right] \\ = \frac{4(s+3)}{[(s+3)^2 + 4]^2}$$